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→ Excess → buffer of margin in  
Time → tolerate , flexibility  
→ Sensitive

میں کو آ رہا ہے

Inherent Excess , manufacturing.



✓  
✓  
✓

## Definition of an ERROR.

An Error is a Numerical computation is the difference b/w the actual value of a quantity and its computed (Approximated value). If  $x^*$  represent the computed value of a quantity, the actual value for which is  $x$  the difference  $e = |x - x^*|$  is called the error of approximation.

## Sources of ERRORS.

1) Gross ERROR

2) Rounding  $\approx$

3) Truncation = (Drop) - chopping Error

0.001 of the line screen

1) Gross ERRORS  $\rightarrow$  <sup>Int</sup> Basic Basic assumption

0.7625

The gross Error are either caused by human mistakes or by the computers.

1) Misreading the figures

2) Use of inaccurate mathematical formula (Algorithm)

3) Use of inaccurate data.

## 2) Rounding ERRORS.

The Error introduced by rounding off numbers to a limited number of decimal places is called the rounding error.

exp  $\pi = 3.141592654 = x$   
 $= 3.142 = x^*$

$$e = |x - x^*| = |3.141592654 - 3.142| =$$

$$e = 0.000407346$$



## Rounding Error Rules:

A) if the first discarded digit is less than 5, the previous digit is unchanged; For exp.

$$56.44 \rightarrow 56.4$$

B) if the first discarded digit is greater than 5, the previous digit is increased by 1, For exp.

$$56.46 \rightarrow 56.5$$

C) If the discarded digit is exactly 5, the previous digit is unchanged if it is even and is increased by 1 if it is odd, For exp.

(i)  $56.45 \rightarrow 56.4$  Even

(ii)  $56.75 \rightarrow 56.8$  Odd

## Truncation Errors

The error is due to truncating the series is called truncation error

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!}$$



## Types of ERROR

There are three types of Errors

- 1) Absolute Error (A.E)  $\rightarrow$
- 2) Relative = (R.E)
- 3) Percentage = (P.E)

### i. Absolute ERROR.

The difference b/w exact value and its approximate value is called absolute error.

$$\Delta x = A.E = | \text{Exact} - \text{Approximate} | = | x - x^* |$$

$$\begin{aligned} \text{eg } \pi &= 1.4142 = x \\ &= 1.414 = x^* \end{aligned}$$

$$\Delta x = A.E = | 1.4142 - 1.414 | = 0.0002$$

$$\Delta x = 0.0002$$

### 2) Relative ERROR (R.E)

is the ratio b/w absolute error and Exact Value and it is denoted by 'R.E'

$$R.E = \frac{\Delta x}{x}$$

### Percentage ERROR (P.E).

if the relative error expressed in term of % then it will be a % error

$$P.E = \frac{\Delta x}{x} \times 100\%$$



(4)

Exp Express the no. upto 3 decimal places  
and find the value of  $\Delta x$ ,  $\delta x$ , P.E

$$2.1514$$

Sol

$$x = 2.1514$$

$$x^* = 2.151$$

$$\Delta x = 2.1514 - 2.151 = 0.0004$$

$$\delta x = \frac{\Delta x}{x} = \frac{0.0004}{2.1514} = 0.000185$$

$$P.E = \delta x \times 100\% = 0.000185 \times 100 = 0.0185\%$$

Exp 0.16152

$$x = 0.16152$$

$$x^* = 0.1615$$

$$x^* = 0.162$$

Exp 3.1426

$$x = 3.1426$$

$$x^* = 3.143$$



## Error in computing values of a function of one variable.

Consider  $y = f(x)$  of a function of one variable then absolute error of a fun is

$$\frac{\Delta y}{\Delta x} = f'(x)$$

absolute Error  $\leftarrow \Delta y = f'(x) \Delta x$

where  $\Delta y$  represent absolute error in function and relative error can be defined as

relative Error  $\leftarrow \delta y = \frac{f'(x) \Delta x}{f(x)}$

eg  $y = x^a$  Find A.E & R.E

sol

$$y = x^a$$

Diff. w.r.t. 'x'

$$\frac{\Delta y}{\Delta x} = a x^{a-1}$$

A.E =  $\Delta y = a x^{a-1} \Delta x$

R.E =  $\delta y = \frac{a x^{a-1} \Delta x}{x^a} = a \frac{\Delta x}{x} = a \delta x$

$$\delta y = a \delta x$$

eg  $y = a^x$  (where  $a$  is any constant)

sol

$$y = a^x$$

Diff w.r.t 'x'

$$\frac{\Delta y}{\Delta x} = a^x \ln a$$

A.E

$$\Delta y = a^x \ln a \Delta x$$

$$R.E = \delta y = \frac{\Delta y}{f(x)} = \frac{a^x \ln a \Delta x}{a^x}$$

$$\delta y = \ln a \Delta x$$



Error in computing values of a function of more than one variable

$$y = f(x_1, x_2, \dots, x_n)$$

$$A.E = \Delta y = \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n$$

$$R.E = \delta y = \frac{\frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n}{f(x_1, x_2, \dots, x_n)}$$

Ex Find A.E, R.E, P.E in given Function.

$$u = f(x, y, z) = xy^2z^3$$

where

$$x = 37.1, y = 9.87, z = 6.052$$

$$\Delta x = 0.3, \Delta y = 0.11, \Delta z = 0.016$$

SOL

$$A.E = \Delta y = \frac{\partial (xy^2z^3)}{\partial x} \Delta x + \frac{\partial (xy^2z^3)}{\partial y} \Delta y + \frac{\partial (xy^2z^3)}{\partial z} \Delta z$$

$$\Delta y = y^2z^3 \Delta x + 2xy^2z^3 \Delta y + 3xy^2z^2 \Delta z$$

$$\Delta y = 12689.286 \quad 30698.07264$$

$$R.E = \delta y = \frac{\Delta y}{f(x, y, z)} = \frac{12689.286}{(37.1)(9.87)^2(6.052)^3}$$

$$\delta y = 0.015$$

$$P.E = 0.015 \times 100\%$$

$$P.E = 1.5$$



Ex2  $f(x, y, z) = e^{xy} \cos 2z$  (Values are same)

### Significant Digits.

A Significant Digits is an approximate number is a digit which gives reliable information about size of the no.

#### Rules:

1) Leading zeros are not significant.

Ex.  $0.002125$  has four significant digits.

2) Following zeros that appears after the decimal point are significant.

Ex.  $0.00202570$  has ~~four~~ six sig. digits.

3) Following zeros appear the decimal point may or may not be significant. Per exp.

Ex.  $2025000$  may have four, five, six & seven sig. di.

4) The significant digits in a number do not depend on the position of the decimal point in the no.

Ex.  $12546$  &  $12546$  both contain five sig. digits.



(8)

Q1. P-1 Handoutconsider the function  $f(x) = \sin x$  near  $x=0$ 

- write the Taylor Polynomial  $P_1(x)$ ,  $P_3(x)$ ,  $P_5(x)$
- Graph  $P_1(x)$ ,  $P_3(x)$ ,  $P_5(x)$  &  $f(x)$  over  $[0, 2\pi]$
- compute  $P_1(0.1)$ ,  $P_3(0.1)$ ,  $P_5(0.1)$  and find A.B.R.E

SOL (a) Taylor Polynomial

$$P(x) = f(x_0) + \frac{(x-x_0)f'(x_0)}{1!} + \frac{(x-x_0)^2 f''(x_0)}{2!} + \dots$$

Consider the function  $f(x) = \sin x$  at  $x = x_0 = 0$ .

$$\text{let } f(x) = \sin x = \sin(0) = 0$$

$$f'(x) = \cos x = \cos(0) = 1$$

$$f''(x) = -\sin x = -\sin(0) = 0$$

$$f'''(x) = -\cos x = -\cos(0) = -1$$

$$f^{(4)}(x) = +\sin x = \sin(0) = 0$$

$$f^{(5)}(x) = \cos x = \cos(0) = 1$$

$$P_1(x) = f(x_0) + (x-x_0)f'(x_0)$$

$$P_1(x) = f(0) + (x-0)f'(0)$$

$$P_1(x) = 0 + (x)1$$

$$\boxed{P_1(x) = x} \text{ Ans}$$

$$P_3(x) = f(x_0) + (x-x_0)f'(x_0) + \frac{(x-x_0)^2 f''(x_0)}{2!} + \frac{(x-x_0)^3 f'''(x_0)}{3!}$$

$$= f(0) + (x-0)f'(0) + \frac{(x-0)^2 f''(0)}{2!} + \frac{(x-0)^3 f'''(0)}{3!}$$

$$= 0 + (x)1 + \frac{(x)^2(0)}{2} + \frac{x^3(-1)}{3!}$$

= 0

$$\boxed{P_3(x) = x - \frac{x^3}{6}} \text{ Ans}$$



Similarly

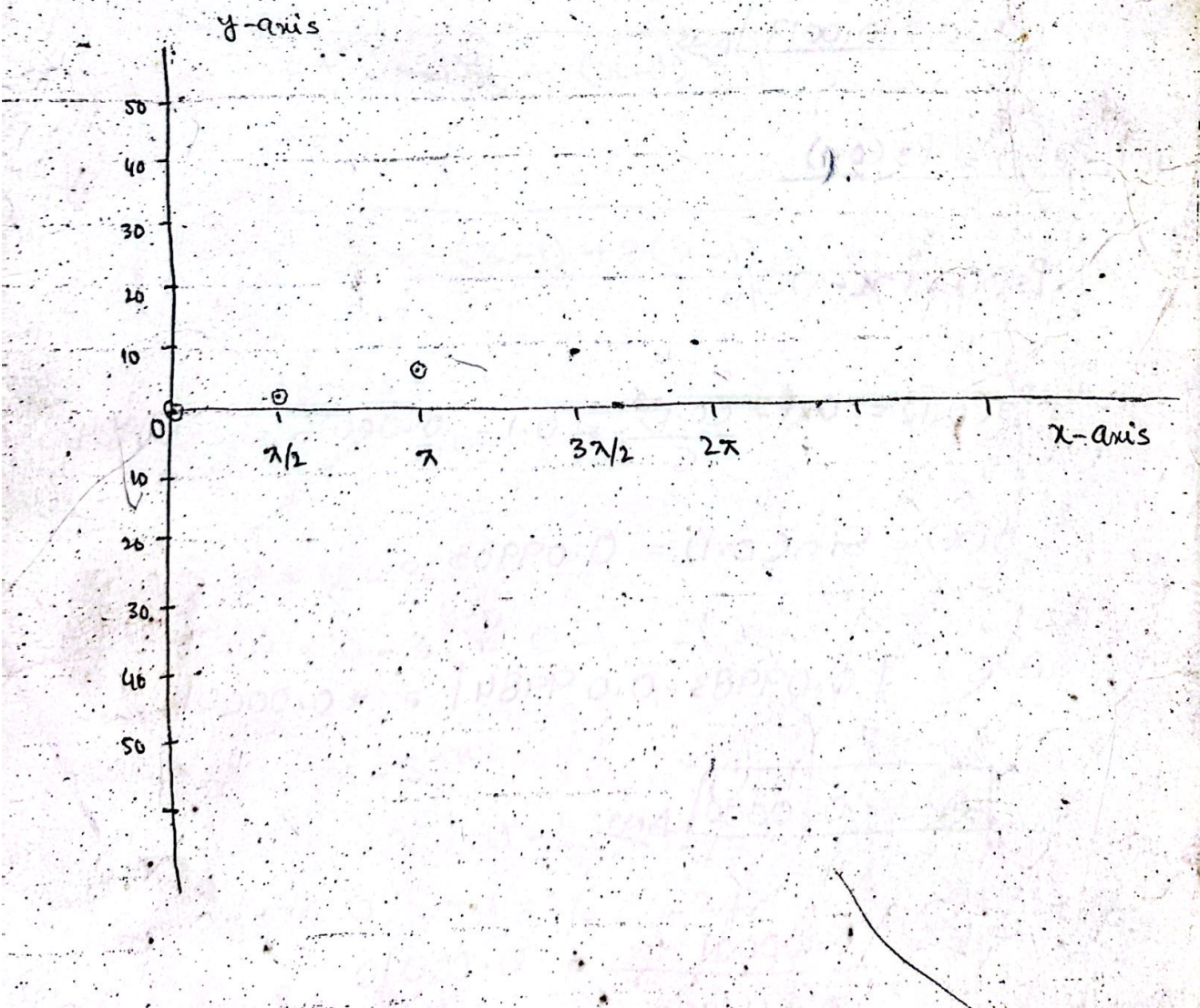
$$P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$$

Axis

(b) Graph

Radian (value calculated)

| $x$      | $P_1(x) = x$ | $P_3(x) = x - \frac{x^3}{6}$ | $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ |
|----------|--------------|------------------------------|------------------------------------------------|
| 0        | 0            | 0                            | 0                                              |
| $\pi/2$  | 1.57         | <del>3.14</del> 0.92         |                                                |
| $\pi$    | 3.14         | -2.01                        |                                                |
| $3\pi/2$ | 4.71         | -12.70                       |                                                |
| $2\pi$   | 6.28         | -33.52                       |                                                |





©

$$P_1(0.1)$$

$$P_1(x) = x$$

$$P_1(0.1) = 0.1 \quad \text{Approx.}$$

$$f(x) = \sin x = \sin(0.1) = 0.09983 \quad (R) \quad \text{Actual value}$$

$$A.E = | \text{Actual value} - \text{Approx. value} |$$

$$A.E = | 0.09983 - 0.1 | = 0.00017$$

$$A.E = 0.00017 \quad \text{Ans}$$

$$R.E = \frac{A.E}{\text{Actual value}} = \frac{0.00017}{0.09983} = 0.0017$$

$$R.E = 0.0017 \quad \text{Ans}$$

$$P_3(x) = P_3(0.1)$$

$$P_3(x) = x - \frac{x^3}{6}$$

$$P_3(0.1) = 0.1 - \frac{(0.1)^3}{6} = 0.1 - 0.00016 = 0.09984$$

$$f(x) = \sin(0.1) = 0.09983$$

$$A.E = | 0.09983 - 0.09984 | = 0.00001$$

$$A.E = 0.00001 \quad \text{Ans}$$

$$R.E = \frac{0.00001}{0.09983} = 0.00010$$



Q4 Find Taylor Polynomial order 2, order 3 at  $x_0 = 1$   
 $f(x) = x^3 + x$

$$f(x) = x^3 + x = f(1) = 2$$

$$f'(x) = 3x^2 + 1 = f'(1) = 4$$

$$f''(x) = 6x = f''(1) = 6$$

$$f'''(x) = 6 = f'''(1) = 6$$

$$P(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2}{2!}f''(x_0) + \dots$$

$$P_2(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2}{2!}f''(x_0)$$

$$P_2(x) = f(1) + (x - 1)f'(1) + \frac{(x - 1)^2}{2!}f''(1)$$

$$P_2(x) = 2 + (x - 1)(4) + \frac{(x - 1)^2}{2!}6$$

$$P_2(x) = 2 + (x - 1)4 + (x - 1)^2 3$$

Similarly

$$P_3(x) = 2 + 4(x - 1) + 3(x - 1)^2 + (x - 1)^3$$

Q5 Find interval.

$$(a) \ x - 3^{-x} = 0$$

$$f(a) = f(x) = x - 3^{-x}$$

$$= f(0) = 0 - 3^0 = 0 - 1 = -1$$

$$f(b) = f(x) = x - 3^{-x}$$

$$= f(1) = 1 - 3^{-1} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$f(a)f(b) < 0 = -1 \times \frac{2}{3} = -\frac{2}{3}$$

Interval  $[0, 1]$  Ans



Q6 (a)  $12.3 + 0.0234$  (3 digit)

Exactly =  $12.3 + 0.0234 = 12.3234$

Rounding =  $12.3$

Chopping =  $12.3$

(b)  $-0.0321 + 0.000136$

Exactly =  $-0.031964$

Rounding =  $-0.032$

chopping =  $-0.031$

(c)  $12.3 - 0.0234$

Exactly =  $12.2766$

Rounding =  $12.3$

chopping =  $12.2$

(e)  $132 \times 0.987$

Exactly =  $130.284$

Rounding =  $130$

chopping =  $130$

---

$(2.75 \times 1.07 - 2.95) 1.07 + 3.16) 1.07 - 4.67$

$(-0.01) 1.07 + 3.16) 1.07 - 4.67$

$(-0.010 + 3.16) 1.07 - 4.67$

$3.37 - 4.67$

$-1.30$  Ans



X

Q7  $2.75x^3 - 2.95x^2 + 3.16x - 4.67$  at  $x = 1.07$   
Find A.E, R.E after chopping.

Sol (a) Proceeding from left to right

$$2.75x^3 - 2.95x^2 + 3.16x - 4.67$$

$$2.75 \times x \times x \times x - 2.95 \times x \times x + 3.16 \times x - 4.67$$

$$2.75 \times 1.07 \times x \times x - 2.95 \times 1.07 \times x + 3.16 \times 1.07 - 4.67$$

After chop  $2.94 \times x \times x - 3.15 \times x + 3.38 - 4.67$

$$2.94 \times 1.07 \times x - 3.15 \times 1.07 - 1.29$$

After chop  $3.14 \times x - 3.37 - 1.29$

$$3.14 \times 1.07 - 4.66$$

After chop  $3.35 - 4.66$

(b)  $-1.33$  Ans

$-1.31$  Ans Approx.

Actual value =  $2.75(1.07)^3 - 2.95(1.07)^2 + 3.16(1.07) - 4.67$

A.V =  $-1.298$

A.E  $|-1.298 + 1.31| = 0.012$  Ans

R.E =  $\frac{0.012}{-1.298} = -0.0092$  Ans

©  $((2.75x - 2.95)x + 3.16)x - 4.67$

Sol

~~$(2.75 \times 1.07 - 2.95) \times x + 3.16 \times x - 4.67$~~

~~$(2.75 \times 1.07 - 2.95) \times (2.75 \times 1.07 - 2.95) \times x + 3.16 \times x - 4.67$~~

~~$((-0.01) \times 1.07 + 3.16) \times x - 4.67$~~

~~$((-0.01 \times 1.07 + 3.16) \times 1.07 - 4.67)$~~

~~$(-0.01 \times 1.07 + 3.16) \times 1.07 - 4.67$~~

~~$3.15 \times 1.07 - 4.67 = -1.30$~~



(14)

X

Q8 write in nested form  $5x^6 + x^5 + 3x^4 + 3x^3 - x^2 + 1$ 

Sol

$$5x^6 + x^5 + 3x^4 + 3x^3 - x^2 + 1$$

$$5x^4 + x^3 + 3x^2 + 3x - 1) x^2 + 1$$

$$5x^3 + x^2 + 3x + 3) x - 1) x^2 + 1$$

$$5x^2 + x + 3) x + 3) x - 1) x^2 + 1$$

$$((((5x+1)x+3)x+3)x-1)x^2+1 \quad \underline{\text{Ans}}$$

↶

Reverse

$$(5x^2 + x + 3)x + 3) x - 1) x^2 + 1$$

$$(5x^3 + x^2 + 3x + 3)x - 1) x^2 + 1$$

$$5x^4 + x^3 + 3x^2 + 3x - 1) x^2 + 1$$

$$5x^6 + x^5 + 3x^4 + 3x^3 - x^2 + 1 \quad \underline{\text{Ans}}$$

Q11

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$(a) S_1 = \sum_{n=0}^5 \frac{1}{n!}$$

$$= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!}$$

$$= \frac{1 + 1 + 0.5 + 0.166\bar{6} + 0.041\bar{6} + 0.008\bar{3}}{2 + 0.5 + 0.166\bar{6} + 0.041\bar{6} + 0.008\bar{3}}$$

$$\frac{2.5 + 0.166\bar{6} + 0.041\bar{6} + 0.008\bar{3}}{2.66 + 0.041\bar{6} + 0.008\bar{3}}$$

$$\frac{2.66 + 0.041\bar{6} + 0.008\bar{3}}{2.70 + 0.008\bar{3}}$$

$$\frac{2.70 + 0.008\bar{3}}{2.70 \text{ Approx}}$$

2.70 Approx

$$e = 2.7182818$$

$$A.E = |2.7182818 - 2.70| = 0.0182 \quad \underline{\text{Ans}}^{818}$$

$$R.E = \frac{0.0182}{2.7182818} = 0.0067 \quad \underline{\text{Ans}}^{254984}$$



$$(b) = S_2 = \sum_{n=0}^5 \frac{1}{(5-n)!}$$

$$= \frac{1}{5!} + \frac{1}{4!} + \frac{1}{3!} + \frac{1}{2!} + \frac{1}{1!} + \frac{1}{0!}$$

$$= \frac{0.008\cancel{8} + 0.041\cancel{6} + 0.166\cancel{6} + 0.5 + 1 + 1}{}$$

$$= \frac{0.049 + 0.166\cancel{6} + 0.5 + 1 + 1}{}$$

$$\frac{0.215 + 0.5 + 1 + 1}{}$$

$$\frac{0.715 + 1 + 1}{}$$

$$\frac{1.715 + 1}{}$$

$$2.71 \text{ Approx.}$$

$$e = 2.7182818$$

$$A.E = |2.7182818 - 2.71|$$

$$A.E = 0.0082 \text{ Ans}$$

$$R.E = \frac{0.0082}{2.7182818} = 0.0030 \text{ Ans}$$

$$\text{Exp } \int_0^{1/2} e^{x^2} dx = 0.544987104184 \quad \text{Pin } (P_8(x)) = ??$$

Compare the N's with  
True Soln

$$\text{Sol } e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\text{Let } e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!} = P_8(x) \rightarrow \text{Derive by Taylor}$$

$$\int_0^{1/2} (1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!}) dx$$

$$= \left| x + \frac{x^3}{3(1!)} + \frac{x^5}{5(2!)} + \frac{x^7}{7(3!)} + \frac{x^9}{9(4!)} \right|_0^{1/2}$$

$$= 0.5447 \text{ Ans}$$



6

## Assignment #1

- 1) Binary Numbers with exs.
- 2) Propagation of Errors with exs.
- 3) Ex on Page 39 Q 1 - ① 10
- 4) Significant Digits

Section C 7/8/06      Submitted Date 11/8/06 Even.  
12/8/06 Odd.

Section B 9/8/06      Submitted Date 16 Aug.

Section A      =      =      =      =

Section D      =      =      =      =